2025 IEEE International Conference on High Performance Computing and Communications (HPCC) | 979-8-3315-6874-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/HPCC67675.2025.00214

Retrieval-Augmented LLM-Driven Multi-Agent Optimization Framework for Intelligent Manufacturing Scheduling

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***Abstract*—Optimizing production scheduling in complex manufacturing environments is essential for improving efficiency and maximizing resource utilization. However, in some situations, traditional optimization methods and current large language models (LLMs) have limitations. We propose a Retrieval-Augmented LLM-driven Multi-Agent Optimization (RALMAO) framework to overcome these shortcomings. This novel architecture integrates Retrieval-Augmented Generation with LLM-driven multi-agent systems to incorporate domain-specific knowledge, which enables inter-agent collaboration and cross-system optimization in manufacturing ecosystems. Validation using real-world industrial datasets—including eye drop and apparel manufacturing—demonstrates the framework's superior performance compared to current enterprise solutions. Our study facilitates the practical application of LLMs in complex scheduling problems by providing actionable insights, which is conducive to improving the intelligent level of enterprises.**

Keywords—Large language models, Retrieval-Augmented Generation, Multi-Agent system, Scheduling problem

I. Introduction

Driven by the digitalization of enterprises, production scheduling—an essential operational function—substantially enhances resource utilization, mitigates production cycle durations, and improves responsiveness to dynamic manufacturing environments [1]. As enterprises scale and market complexity intensifies, production scheduling has evolved from basic task allocation into a multifaceted challenge involving both resource optimization and production scheduling. This function interlinks tightly with inventory management, order processing, and quality control systems to maximize manufacturing efficiency and resource utilization [2]. Nevertheless, despite deploying Advanced Planning and Scheduling (APS) systems, many factories experience suboptimal scheduling outcomes due to complex production constraints [3]. While manual scheduling leverages experiential knowledge to partially address complexity, its inefficiency renders it inadequate for large-scale production tasks [4]

To address these challenges, traditional optimization methods (e.g., mathematical programming, heuristic and metaheuristic algorithms [5], [6], [7], [8]) and emerging deep learning and reinforcement learning (RL) approaches [9], [10] have been widely adopted. However, these techniques exhibit inherent limitations due to their dependence on expert-designed training strategies. Large language models (LLMs) have advanced natural language processing capabilities that provide new opportunities by accurately interpreting textual scheduling requirements [11], [12]. Multi-agent systems (MAS) further enable collaborative LLM agents to address distinct scheduling dimensions [13].

However, current LLM-based methods face critical constraints: rigid architectures limit adaptability, poor cross-system coordination hinders optimality, and high domain knowledge transfer costs restrict scalability. These gaps necessitate a flexible, coordinative, and knowledge-efficient framework. To overcome these barriers, we propose the Retrieval-Augmented LLM-Driven Multi-Agent Optimization (RALMAO) framework for complex production scheduling.

By integrating Retrieval-Augmented Generation (RAG) knowledge bases [14] with seamless agent coordination, our framework dynamically adjusts scheduling plans in response to real-time production fluctuations, enabling efficient and precise scheduling.

Our work makes three primary contributions. First, we establish an innovative and extensible platform to accommodate diverse production scheduling requirements across industries. Second, we achieve effective domain knowledge infusion and cross-system decision optimization in enterprises, reducing knowledge transfer costs while enhancing system adaptability. Finally, we validate the

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framework's effectiveness in enhancing scheduling efficiency using real data from two industrial domains.

This framework provides an extensible optimization scheduling tool under resource allocation constraints, which extends LLM applications in production optimization and industrial digitalization. The following section reviews existing traditional methods, LLM-based methods, RAG techniques, and MAS for scheduling optimization, establishing the rationale for our novel framework. First, Section III describes our primary contributions and framework components: beginning with an architectural overview, followed by comprehensive component explanations. Then, Section IV evaluates the framework using real-world datasets. Finally, Section V summarizes the research contributions, discusses their implications for scheduling optimization, and outlines future research directions.

II. Related Works

A. Optimization Methods for Scheduling

Various methods are adopted in the scheduling optimization field to resolve problems with varying complexity and requirements. Traditional optimizations include mathematical programming, heuristic methods, and metaheuristic algorithms. Research [6] investigating flexible job-shop integrated planning and scheduling established a mixed-integer programming model capable of rapidly obtaining near-optimal solutions. However, its computational complexity escalates exponentially with problem scale, constraining applicability in large-scale scenarios. Heuristic methods emerged to overcome this limitation, including greedy algorithms and nearest neighbor approaches. Researchers developed a greedy algorithm-based heuristic in supermarket scheduling contexts that generates feasible schedules rapidly through expedited task assignment and conflict resolution. However, solution quality may be compromised by local optima [15]. Metaheuristic techniques such as genetic algorithms (GA), particle swarm optimization (PSO), and ant colony optimization (ACO) explore solutions across wider search spaces by simulating natural evolution or collective behaviors, rendering them suitable for complex, high-dimensional scheduling problems. Nevertheless, these methods exhibit parameter sensitivity and require expert-guided tuning to ensure high-quality solutions [7].

With advancements in artificial intelligence (AI), machine learning approaches have progressively been used in scheduling optimization. For instance, Zang et al. proposed a hybrid deep neural network scheduler combining convolutional and backpropagation networks. This model decomposes job-shop scheduling problem (JSSP) instances into subproblems based on job processing sequences and machine priorities by extracting features from genetic algorithm-generated training samples. However, the method demands substantial high-quality training data and incurs prolonged training durations [16]. RL, wherein agents learn optimal policies through environmental interactions, demonstrates significant advantages for complex scheduling scenarios[17]. When addressing integrated process planning and scheduling (IPPS) problems, researchers developed a deep reinforcement learning (DRL)-based approach that substantially enhances solution efficiency and quality in large-scale IPPS instances. However, computational costs during training and sensitivity to environmental dynamics remain considerations [18].

In summary, while these methods offer distinct advantages for scheduling problems, they still have common limitations: dependence on experts to design problem-specific models and algorithms, necessitating deep domain understanding and extensive experience. Furthermore, scheduling problems frequently intertwine with inventory management, order processing, and other operational workflows in dynamic industrial environments such as eye drop manufacturing. This complexity demands scheduling optimization techniques capable of dynamic strategy adaptation [19]. Consequently, more intelligent optimization mechanisms are required for systems that integrate domain expertise (e.g., material formulations, production prioritization) while enabling seamless interoperability with manufacturing execution systems (MES), warehouse management systems (WMS), and other platforms to dynamically adjust optimization strategies in response to multifaceted industrial scheduling scenarios.

B. Large Language Model (LLM)-driven Scheduling

LLMs refer to massively parameterized models (e.g., GPT series) pre-trained on extensive datasets, exhibiting natural language understanding and generation capabilities [20]. Their core value in industrial scheduling lies in transforming unstructured constraints described in natural language into computable mathematical logic [11], [12], [13]. RAG constitutes a framework integrating retrieval and generation mechanisms. RAG mitigates hallucination risks in industrial contexts by incorporating domain knowledge through real-time retrieval from external knowledge bases (e.g., equipment manuals, GMP specifications) into the model's context, ensuring generated content adheres to professional rules [14].

Due to their robust natural language processing and reasoning capacities, LLMs have appeared promising tools for solving optimization problems. One study demonstrated the efficacy of Optimization by Prompting (OPRO) in addressing linear regression problems and Traveling Salesman Problem (TSP) instances [11]. Another approach uses LLMs as mutation operators in continuous multi-objective optimization algorithms, exemplified by Liu et al.'s LLM-driven Evolutionary Algorithm (LMEA), which leverages LLMs to generate novel candidate solutions, thereby advancing multi-objective optimization methodologies [12]. Nevertheless, these methods face significant limitations: insufficient utilization of LLMs' reasoning capabilities for complex scheduling constraints; performance degradation and critical information loss when single LLM models process excessively lengthy prompts; and difficulties in accurately capturing constraint quantities and complexity. RAG effectively solved these challenges by retrieving relevant knowledge to enhance the model's problem comprehension and reasoning capacity, ensuring precise capture and utilization of critical information when handling intricate scheduling constraints, consequently improving model performance in optimization tasks [14].

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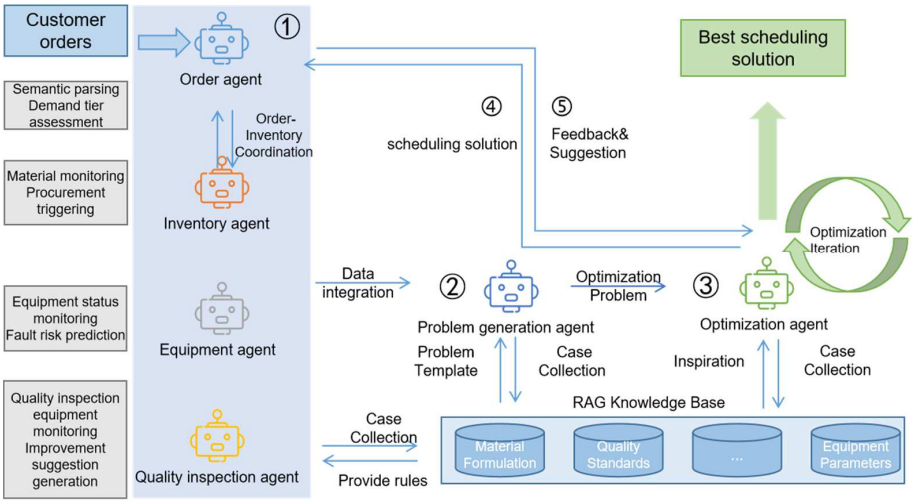


Fig. 1. The RALMAO Framework Overview

LLM-driven MAS comprises multiple autonomous agents proficient in completing tasks through collaboration or competition within dynamic environments. For instance, one study constructed an LLM-based multi-agent meta-programming framework that decomposes complex tasks by assigning specialized roles (e.g., product manager, architect, engineer), efficiently handling sophisticated development assignments [21]. Within scheduling optimization—as an emerging frontier—researchers have developed LLM-driven multi-agent evolutionary frameworks to address diverse optimization problems [13], [22].

However, in complex industrial scenarios, these approaches exhibit three persistent shortcomings: First, the absence of dynamic adjustment mechanisms—once optimization model types are configured, online adaptation becomes infeasible; Second, inadequate cross-system coordination—industrial safety-critical contexts (e.g., ophthalmic solution production) necessitate deep coupling between scheduling systems and MES or WMS, requiring synchronous incorporation of expert knowledge such as raw material expiration dates and formulations; Third, prohibitive knowledge transfer costs—domain transitions (e.g., semiconductor to pharmaceutical manufacturing) demand reconstruction of optimization logic due to deficient generalizable reasoning capabilities. These deficiencies underscore the imperative for dynamic optimization mechanisms with knowledge injection—integrating domain-expert rules with real-time system feedback—to achieve multi-objective collaborative optimization strategies within industrial multi-agent scheduling applications.

III. Methodology

A. Framework overview

The RALMAO framework, illustrated in Fig. 1, represents a cross-industry production scheduling solution for complex manufacturing scenarios. This architecture integrates RAG technology with LLM-driven MAS to achieve domain-specific expertise injection and cross-system decision optimization within enterprises. The framework coordinates six specialized agents: order agent, inventory agent, equipment agent, quality inspection agent, problem formulation agent, and optimization agent, facilitating inter-agent information sharing and knowledge infusion through a unified RAG knowledge base. Ultimately, through an iterative optimization mechanism, it continuously performs data collection, problem definition, solution generation, evaluation feedback, and strategy adjustment. Each iteration improves based on previous results and feedback until preset optimization objectives or convergence conditions are met, achieving continuous production scheduling optimization. The pseudocode of the proposed framework is provided below.

Algorithm 1: Pseudocode of Retrieval-Augmented LLM-Driven Multi-Agent Optimization (RALMAO) framework

Input: customer orders C, max agent iterations M, max optimization iterations G, fitness threshold ε
Output: best scheduling solution S^*

Initialize RAG knowledge base;
For i = 1 to M do
  order requirements ← order agent
  inventory status ← inventory agent
  equipment status ← equipment agent
  quality status ← quality inspection agent
  optimization problem ← problem generation agent
    For j = 1 to G do
    S_{ij} ← optimization agent.GA
    fitness ← calculate fitness
      if fitness < ε then
        break
    End For
   agent feedback ← S_{ij} is distributed to the agents
  if agent feedback satisfies requirements, then
    S^* ← S_{ij}
    break
adjust S_{ij} according to agent feedback
End For

The order agent is responsible for semantically understanding order requirements, transforming customer inputs into production plans, and dynamically responding to changes; the inventory agent manages inventories of raw materials, auxiliary materials, and packaging materials, utilizing the RAG knowledge base for demand calculation and inventory optimization; the equipment agent monitors

This work was supported by the Foundation of China Unicom under Grant V91D241DZH0000.

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equipment status, predicts potential failures, and optimizes maintenance plans; the quality inspection agent monitors the status of quality inspection equipment, and provides quality improvement suggestions; the problem generation agent extracts content generated by order, inventory, equipment, and quality inspection agents, filters key parameters, and combines them to define problems in production scheduling (e.g., constraints, objectives); the optimization agent employs optimization algorithms such as GA and RL methods to dynamically generate scheduling solutions. The RAG knowledge base is the foundational component, storing critical information such as product formulations, raw material requirements, and constraint rules to ensure the efficient and accurate operation of the scheduling system.

B. RALMAO Framework

1) Agent Design

a) Order agent: processes order requirements through semantic understanding, converting customer inputs into actionable production plans. It extracts critical information (e.g., delivery timelines, priorities, material specifications) from order data to facilitate production scheduling. Leveraging LLM capabilities, the agent parses orders containing complex textual specifications and employs natural language processing to identify urgency levels and commercial value, thereby enabling prioritized resource allocation.

b) Inventory agent: manages and monitors inventories of raw materials, auxiliary supplies, and packaging materials. Utilizing the RAG knowledge base, it interprets product formulation rules and calculates material requirements to conduct real-time inventory assessments. Beyond routine stock inspections, the agent forecasts inventory needs based on production plan fluctuations and market trends, initiating replenishment or inventory reduction actions.

c) Equipment agent: monitors production equipment status and provides maintenance recommendations. It tracks real-time operational metrics (e.g., runtime, fault detection) via integrated sensor data to ensure equipment stability while feeding status updates to the problem generation agent for scheduling optimization. Additionally, the agent predicts potential failures using historical maintenance data, develops preventive maintenance schedules, and mitigates unplanned downtime impacts.

d) Quality inspection agent: monitors quality control equipment by analyzing real-time operational data from integrated sensors. It immediately coordinates with other agents to issue alerts upon detecting equipment anomalies or non-compliant materials/products, enabling prompt corrective interventions. Relevant quality data is concurrently relayed to the Problem Generation Agent.

e) Problem generation agent: synthesizes production data, inventory status, and order requirements to generate structured datasets and optimization objectives. It identifies production bottlenecks through historical data analysis, providing improvement recommendations to enhance scheduling efficiency.

f)Optimization agent: generates production schedules meeting specified objectives using structured inputs from the Problem Generation Agent. It employs multiple optimization methods—including GA and RL—to ensure solutions satisfy targets within predefined iteration limits.

The specific implementation of genetic algorithms within this agent follows these steps:

- Chromosome encoding: In the genetic algorithm, each production plan is encoded as a chromosome, with each gene representing a production task or decision variable containing information such as task priority, equipment allocation, and material requirements.

$$C = [g_1, g_2, .., g_n] \tag{1}$$

where g_i denotes the i-th production task, encompassing attributes like task identifier, equipment requirements, and material demand.

- Population initialization: The population is initialized by randomly generating a set of chromosomes, each representing a potential production plan. Problem complexity typically determines population size, ranging between 20 and 200.

- Fitness function: The fitness function evaluates the quality of each chromosome (production plan). For eye drop scheduling, it incorporates multiple objectives: production efficiency, inventory costs, and changeover time. The fitness function is defined as:

$$Fitness = w_1 \times Efficiency + w_2 \times {}^{1}/_{Inventory\ Cost} + w_3 \times {}^{1}/_{Changover\ Time} \tag{2}$$

where w_1, w_2, w_3 are weight coefficients adjusted per operational requirements; *Efficiency* represents production efficiency (measured by completed tasks/time); *InventoryCost* denotes inventory cost (calculated from inventory levels and holding costs); *ChangeoverTime* indicates changeover time (measured by equipment switching duration).

- Selection: Selection operators choose higher-fitness individuals from the current population for the mating pool to produce offspring. Standard methods include the Roulette Wheel Selection and Tournament Selection. The roulette wheel selection probability is:

$$P_i = \frac{Fitness_i}{\sum_{j=1}^{N} Fitness_j} \tag{3}$$

where P_i is the selection probability of the i-th individual, and N is the population size

- Crossover: Crossover combines two parent production plans to generate offspring. Single-point crossover selects a random position p, splits the parent plans at this point, and exchanges segments to create offspring.

$$C_{offspring} = \begin{cases} C_1[0:p] + C_2[p:], & P_c \\ C_2[0:p] + C_1[p:], & 1\text{-}P_c \end{cases} \begin{cases} C_1[0:p] + C_2[p:], & P_c \\ C_2[0:p] + C_1[p:], & 1\text{-}P_c \end{cases} \tag{4}$$

where p is the randomly selected crossover point, and P_c (crossover probability) typically ranges from 0.6 to 0.9.

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- Mutation: Mutation randomly alters genes to increase population diversity and prevent local optima. Exchange mutation randomly selects two positions i and j and exchanges the genes at these two positions.

$$C_{mutated} = \begin{cases} C[0:i] + [g_j] + C[i+1:j] + [g_i] + C[j+1:], & P_m \\ C, & 1\text{-}P_m \end{cases} \quad (5)$$

where P_m is the probability of mutation, generally ranging from 0.01 to 0.1.

- Iterative optimization: The algorithm iteratively optimizes the population through these steps: calculate individual fitness; select high-fitness individuals for mating pool; perform crossover and mutation to generate offspring; merge parent and offspring population; select highest-fitness individuals as the new generation; termination occurs upon reaching maximum iterations or convergence (e.g., fitness plateau).
- Result Evaluation: At the end of the iteration, the highest-fitness individual is selected as the optimal production plan. Output includes task sequence, equipment allocation, and material requirements for production guidance.

The generated solutions are distributed to agents for evaluation. Each agent employs its domain-specific knowledge to assess the plans and propose improvement suggestions. The optimization agent dynamically adjusts the scheduling solutions based on feedback. This process iteratively refines the optimization results until the optimization objectives are met, or the convergence criteria are satisfied.

2) RAG Knowledge Base

The RAG knowledge base plays a crucial role in the scheduling framework by providing accurate and reliable optimization information to facilitate agent-based decision-making. Specifically, it functions as a repository for product formulations, material/packaging requirements, constraint rules, and historical strategies. Additionally, it enhances data privacy.

IV. Experiments

To demonstrate the effectiveness of the proposed framework, we conducted experiments using real-world data from two industries. The framework's performance was compared with the manual scheduling methods currently utilized by enterprises.

TABLE I. Production Environment

Parameter	*Value*	*Description*
Number of production lines	3	One each for A/B/C cleanroom levels
Product types	5	Includes antibiotics/artificial tears/anti-allergy
Daily output	2 batches	Quantity per batch is in the knowledge base
Probability of emergency order	10%	Highest priority

The first experiment used one month's order data from an eye drop factory, the production environment is detailed in TABLE I. The optimization agent uses GA method, the vector database of the knowledge base is constructed using Milvus and contains basic information such as raw materials, auxiliary materials, packaging materials, product formulas, and constraint rules for each eye drop type. The Chinese version of BERT-based Generalized Embedding is used as the embedding model, and the large language model employed for enhanced generation and agents is DeepSeek-R1: 32b. As shown in TABLE II, compared to the manual scheduling methods currently used by enterprises, the RALMAO framework has effectively shortened the changeover time and urgent order response time, improved production efficiency and the number of early warnings for material shortages, verifying the effectiveness of the RALMAO framework.

the average changeover time was reduced by 52%, production efficiency increased by 38%, the average response time for urgent order insertion was less than 8 minutes, and there were 5 material shortage warnings, confirming the effectiveness of the RALMAO framework.

TABLE II. Performance Comparison Results for Experiment 1

Metric	*Manual Scheduling*	*RALMAO Framework*	*Improvement*
Average Changeover Time (min)	78	37	↓ 52%
Production Efficiency	65%	90%	↑ 38%
Inventory Turnover Rate (times/month)	2.1	3.8	↑ 81%
Urgent Order Response Time (min)	45	8	↓ 82%
Material Shortage Warnings (times/month)	2	5	↑ 150%

The second experiment utilized data from the apparel industry's hanging line human-machine scheduling to test the framework's capability for human-machine collaborative optimization in multi-category mixed-flow scenarios. We combined GA with RL for optimization agent selection, automatically generating optimal workstation allocation plans and supporting rapid rescheduling for abnormal events (such as equipment failures) As shown in TABLE III, compared to the currently used methods, real-time optimization of the scheduling plan reduced production response time from hours to under 5 minutes; equipment utilization increased by over 30%, and production line changeover efficiency improved by over 40%.

TABLE III. Performance Comparison Results for Experiment 2

Metric	*Manual Scheduling*	*RALMAO Framework*	*Improvement*
Rescheduling Response Time	2.5h	4.2min	↓ 97.2%
Equipment Utilization	62%	81%	↑ 30.6%
Production Line Changeover Efficiency	Baseline	42%	↑ 42%

Compared to manual methods, the results demonstrate that the RALMAO framework has significant advantages in enhancing production scheduling efficiency and response speed in both the eye drop and apparel manufacturing

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scenes. It effectively addresses the production scheduling needs across different industries. However, it is important to note that the primary aim of this framework is to showcase the ability of RAG technology with LLM-driven MAS to coordinate inter-system resources and optimize scheduling problems through collaborative interaction. While its absolute optimization performance for specific problems may not match that of dedicated solvers (such as CPLEX for solving MILP), its core value lies in cross-industry adaptability, dynamic response capability, and expert knowledge transfer efficiency, providing a scalable solution for intelligent manufacturing scheduling.

V. Discussions and Conclusion

This study proposes an innovative RALMAO framework that optimizes production scheduling in complex industrial environments. The framework achieves cross-system decision optimization and feedback iteration by integrating MAF with RAG knowledge bases. Validation on real-world datasets from the eye drop and apparel industries demonstrates significant superiority over traditional manual scheduling methods. The framework reduces scheduling duration from hours to minutes, decreases production delays and equipment idle time, and improves equipment utilization, confirming its efficiency and practicality in real production environments.

The framework provides a flexible and extensible tool that significantly enhances LLMs' capabilities in understanding domain-specific knowledge, reasoning, and rapid updating. It demonstrates the potential and advantages of large models in solving complex production scheduling problems, serving as a case study for applying AI to practical production scheduling.

Future work will focus on expanding the framework's applicability to additional sectors including food processing and electronics manufacturing to validate its universality and portability, with customized adaptations for industry-specific requirements; exploring integration with emerging technologies such as digital twins and IoT to enable precise real-time monitoring and predictive maintenance, thereby enhancing dynamic adaptability and intelligent capabilities; and investigating multi-factory collaborative scheduling through federated learning architectures.

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